REQUIREMENTS SPECIFICATION

University Lost and Found System

A centralized platform to report, match, verify and return lost and found items on campus

Project	**University Lost and Found System** (proposed product name: Retrouv'UNI)
Context	University environment (students, staff, lost and found office)
Document	Software Requirements Specification
Version	1.0
Date	October 2026
Status	Draft for review

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1. Background and Problem Statement

Since the start of our third year, some of us have lost belongings, or seen others lose them, in classrooms and other campus spaces. A loss is often noticed only later, so nobody knows what happened or whether someone found the item.

Students and staff may lose personal belongings such as phones, bags, keys, student ID cards, documents, books and USB drives. Found items may remain unreported, or be difficult to connect with their owners, because there is **no centralized and organized system**. The same problem likely exists in other universities and schools.

2. Project Objectives

2.1 General objective

Provide a centralized platform where users can report lost or found items, search reports, identify possible matches, verify ownership and record the return of recovered items.

2.2 Specific objectives

Users must be able to:

1. Report an item as **lost**;
2. Report an item as **found**;
3. **Search** reported items;
4. Receive **possible matches** between lost and found reports;
5. Provide the information needed to **verify ownership**;
6. **Follow the status** of their report;
7. Have the **final return recorded**.

2.3 Out of scope

- The system **does not promise to recover stolen items** and does not replace a report to the authorities.
- No payments, no delivery of items, no real-time geolocation.
- No native mobile application in version 1 (responsive web application only).

3. Scope and Main Workflow

The scope is deliberately limited to a single business workflow:

Report → **Search / Match** → **Verify** → **Return** → **Record**

We address the information-management problem: how a university can organize lost and found information and facilitate recovery.

4. Actors and Roles

Actor	Description	Main permissions
Student	Authenticated member of the university	Report lost/found items, search, view matches, claim items, track own reports
University staff	Lecturers, administrative staff, cleaning and security staff	Same as students; can drop found items at the desk
Lost and found officer	Staff member responsible for the lost and found office	Validate reports, manage physical inventory, verify ownership, record returns, view statistics
System administrator	IT team	Manage accounts and roles, categories, locations, settings; view audit logs
Visitor (not logged in)	Anyone	View the home page and the public list (limited information)

5. Functional Requirements

Priorities: **M** = Must (required for v1), **S** = Should (desirable), **C** = Could (optional).

5.1 Authentication and accounts

ID	Requirement	Prio
F-01	Log in with university e-mail (or student ID number) and password	M
F-02	Registration restricted to university domain addresses, with e-mail confirmation	M
F-03	Password reset	M
F-04	Profile management (name, optional phone, faculty/department, notification preferences)	S
F-05	Single sign-on (university SSO / LDAP)	C

5.2 Reporting a lost item

ID	Requirement	Prio
F-10	Form: category, title, description, colour, brand, estimated date and time range, location (building / room), optional photo	M
F-11	**Private** verification fields (distinctive mark, contents, serial number, wallpaper…) visible only to the lost and found officer	M
F-12	Edit or cancel the report as long as it is not closed	M
F-13	Mark as "found it myself" to close the report	S

5.3 Reporting a found item

ID	Requirement	Prio
F-20	Form: category, **deliberately partial** public description, place and date found, photo (blurring sensitive data recommended)	M
F-21	State where the item is: "I am keeping it" or "dropped at desk X"	M
F-22	The officer confirms physical receipt and assigns an **inventory code** (e.g. `LF-2026-0142`)	M
F-23	Identity documents (student ID, national ID card) are flagged as high priority to the officer	S

5.4 Search and browsing

ID	Requirement	Prio
F-30	Full-text search with filters: type (lost/found), category, location, period, status	M
F-31	Paginated list of cards showing photo, category, location, date and status	M
F-32	Detailed report page (without private fields)	M
F-33	Sort by date or relevance	S

5.5 Matching

ID	Requirement	Prio
F-40	For each new report, the system computes a **match score** against reports of the opposite type	M
F-41	Criteria: same category (mandatory), date proximity (± 7 days), same building/location, colour, brand, shared keywords	M
F-42	Matches above a threshold (e.g. 60%) are suggested to the reporter and to the officer	M
F-43	Users can mark a match as "Not my item"	S

5.6 Claims and ownership verification

ID	Requirement	Prio
F-50	The presumed owner **claims** a found item and answers verification questions	M
F-51	The officer compares the answers with the private information and **accepts** or **rejects** the claim (reason mandatory for rejection)	M
F-52	When accepted, a pick-up appointment is proposed (place, time slot)	M
F-53	Several claims on the same item are handled by the officer; only one can be accepted	M

5.7 Return and record keeping

ID	Requirement	Prio
F-60	The officer records the return: date, verified identity (document presented), officer, signature/QR code	M
F-61	The linked lost and found reports move to **Returned** status and are then archived	M
F-62	Searchable history of all reports and returns	M
F-63	Items unclaimed after N days (configurable, e.g. 90): status **Unclaimed** → donation / disposal / handed to administration	S

5.8 Notifications

ID	Requirement	Prio
F-70	In-app notifications (bell): new match, claim received, decision, appointment, return	M
F-71	E-mail notifications	S
F-72	SMS / WhatsApp notifications	C

5.9 Administration and statistics

ID	Requirement	Prio
F-80	Officer dashboard: pending reports, claims to process, items in stock, returns this month	M
F-81	Statistics: locations and periods with the most lost/found items, return rate, average time to return	S
F-82	Management of users, roles, categories, buildings/rooms and desks	M
F-83	CSV/PDF export of registers	S
F-84	Audit log of sensitive actions	M

6. Business Rules

ID	Rule
BR-01	Only an authenticated user can report, claim or receive notifications.
BR-02	A report has **exactly one author** and belongs to **exactly one category**.
BR-03	The date an item was lost/found cannot be in the future.
BR-04	Verification information (private fields) is **never** displayed publicly.
BR-05	A found item can only be returned after a claim has been **accepted** by the officer.
BR-06	A found item can have **only one** return.
BR-07	A user cannot claim an item they reported as found themselves.
BR-08	After 3 rejected claims within 30 days, the account is flagged for review.
BR-09	A report with no activity for 90 days becomes **Expired** (lost) or **Unclaimed** (found).
BR-10	Every return requires presentation of an identity document or student ID card.

Status life cycle

Lost item: Open → Match found → Under verification → Returned → Archived (or Cancelled / Expired)

Found item: Reported → In stock (received at desk) → Claimed → Under verification → Returned → Archived (or Unclaimed)

7. Main Use Cases

UC	Actor	Description
UC-01	Student / Staff	Authenticate
UC-02	Student / Staff	Report a lost item
UC-03	Student / Staff	Report a found item
UC-04	All	Search reports
UC-05	System	Compute and suggest matches
UC-06	Student / Staff	Claim a found item
UC-07	Lost and found officer	Verify ownership
UC-08	Lost and found officer	Record the return
UC-09	Student / Staff	Track the status of own reports
UC-10	Lost and found officer	View statistics
UC-11	System administrator	Manage users, categories and locations

Main success scenario — UC-06 / UC-07 / UC-08

1. The student receives a notification: "Possible match at 87%".
2. They open the report, click "This is my item" and answer the verification questions.
3. The officer compares the answers with the private details and accepts the claim.
4. The student receives a pick-up appointment at the desk.
5. At the desk, the officer checks the student ID, hands over the item and records the return.
6. Both reports move to **Returned**; the history is kept.

8. Conceptual Data Model

Entity	Main attributes
User	id, last_name, first_name, email, student_id, phone, role, department, active, created_at
Category	id, label (Phone, Bag, Keys, Card, Document, Book, USB drive, Clothing, Other), icon
Location	id, building, room/area, campus
Desk	id, name, location_id, opening_hours
Report	id, type (LOST/FOUND), title, description, colour, brand, event_date, location_id, category_id, author_id, status, desk_id, inventory_code, created_at
PrivateDetail	id, report_id, question/hint, value
Photo	id, report_id, url
Match	id, lost_report_id, found_report_id, score, status (SUGGESTED/REJECTED/CONFIRMED)
Claim	id, found_report_id, claimant_id, answers, status (PENDING/ACCEPTED/REJECTED), reason, processed_by, date
Return	id, claim_id, officer_id, document_presented, date, signature
Notification	id, user_id, type, message, read, date
AuditLog	id, user_id, action, entity, entity_id, date

Key relationships: a User has **0..n** Reports; a Report has **0..n** Photos and PrivateDetails; a found Report has **0..n** Claims; an accepted Claim has **0..1** Return.

9. Non-Functional Requirements

ID	Category	Requirement
NF-01	Security	Hashed passwords (bcrypt/argon2), mandatory HTTPS, protection against CSRF, XSS and SQL injection
NF-02	Privacy	Compliance with personal data protection principles; data minimisation; private fields encrypted
NF-03	Access control	Role-based access control (RBAC) enforced server-side
NF-04	Performance	Response time under 2 s for 95% of requests; search under 1 s over 10,000 reports
NF-05	Availability	99% during academic periods; daily database backup
NF-06	Usability	Responsive, mobile-first interface; a report can be completed in under 2 minutes
NF-07	Accessibility	WCAG 2.1 AA contrast, keyboard navigation, labelled form fields
NF-08	Languages	English and French
NF-09	Maintainability	Version-controlled code (Git), layered architecture, automated tests of business rules
NF-10	Traceability	Audit log of approvals, rejections and returns

10. Proposed Technical Architecture

- **Frontend:** responsive web application (HTML/CSS/JavaScript, or React/Vue).
- **Backend:** REST API (Node.js/Express, Laravel or Django, at the team's choice).
- **Database:** relational (MySQL or PostgreSQL).
- **Photo storage:** server folder or object storage, with resized images.
- **Notifications:** SMTP e-mail service; in-app notifications stored in the database.
- **Deployment:** university server or cloud hosting, with a test environment.

```
[Browser]  ⇄  [Web frontend]  ⇄  [REST API]  ⇄  [Database]
                                      │
                                      ├── Matching service (scoring)
                                      ├── Photo storage
                                      └── E-mail / notification service
```

11. Visual Identity (Design System)

Element	Choice
Primary colour	Academic blue `#1E40AF`
Secondary colour	Amber `#F59E0B` ("found" actions, highlights)
Success	`#16A34A` (Returned)
Alert	`#DC2626` (Lost, rejection)
Neutrals	`#0F172A` text · `#64748B` secondary text · `#E2E8F0` borders · `#F8FAFC` background
Typography	**Inter** — headings 600/700, body 400; sizes 12 / 14 / 16 / 20 / 28 / 36 px
Corner radius	8 px (inputs, buttons), 12 px (cards)
Spacing	4 px grid (4, 8, 12, 16, 24, 32, 48)
Icons	Font Awesome
Status badges	Lost (red), Found (amber), Under verification (violet), Returned (green), Archived (grey)

12. User Interface Design

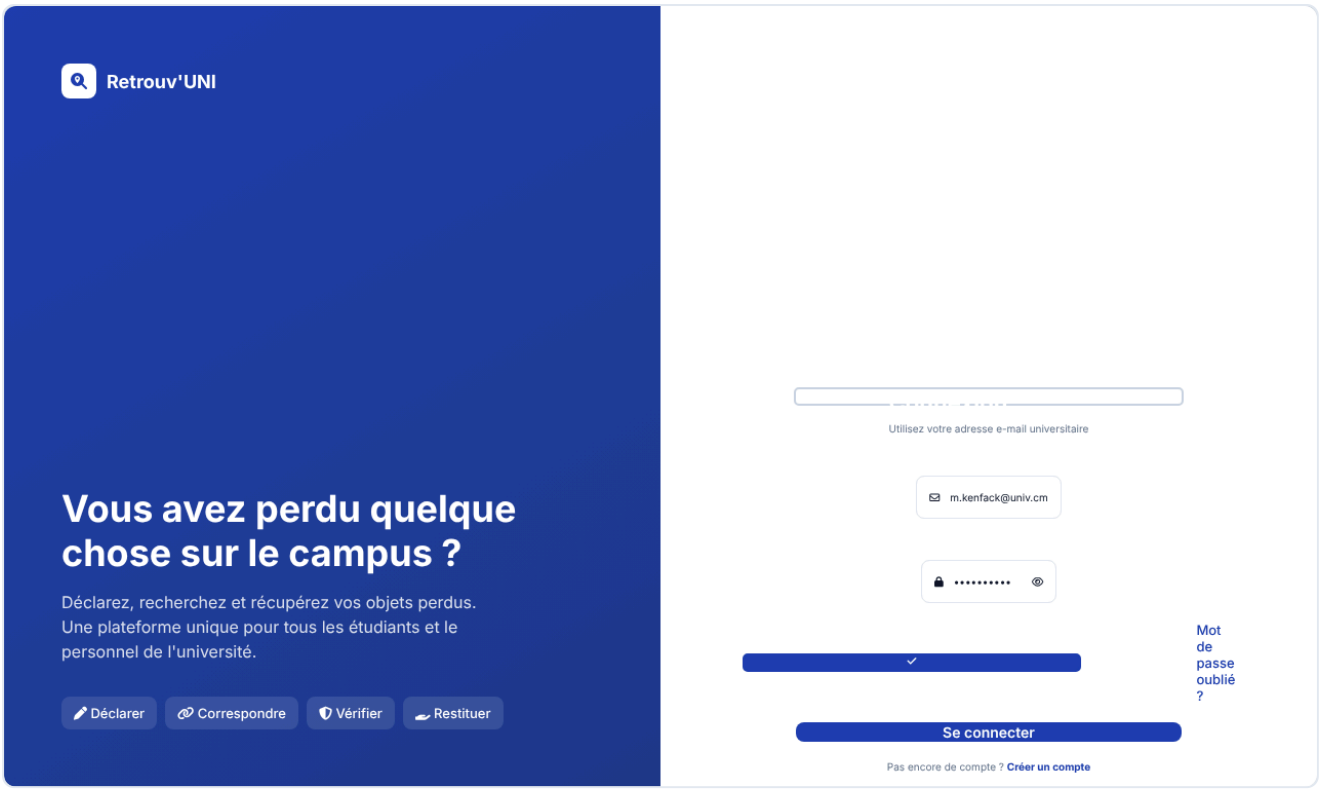
High-fidelity mockups are provided in `maquettes/index.html` and can be imported into Figma as editable layers. Each screen is a 1280 px (desktop) or 390 px (mobile) frame.

#	Screen	Actor	Content
S1	Login	All	Logo, university e-mail, password, sign-up link
S2	Home / Dashboard	Student	"I lost" / "I found" buttons, search bar, matches, my reports
S3	Report a lost item	Student	Step-by-step form, category tiles, location, date, photo, private details
S4	Search	All	Side filters, card grid, status badges
S5	Report details + matches	Student	Item sheet, status timeline, scored matches, "This is my item" button
S6	Ownership verification	Officer	Side-by-side comparison of claimant answers vs private details, accept / reject
S7	Officer dashboard	Officer	KPIs, chart by location, task queue, return recording
S8	Mobile version	Student	Home and reporting on a 390 px screen

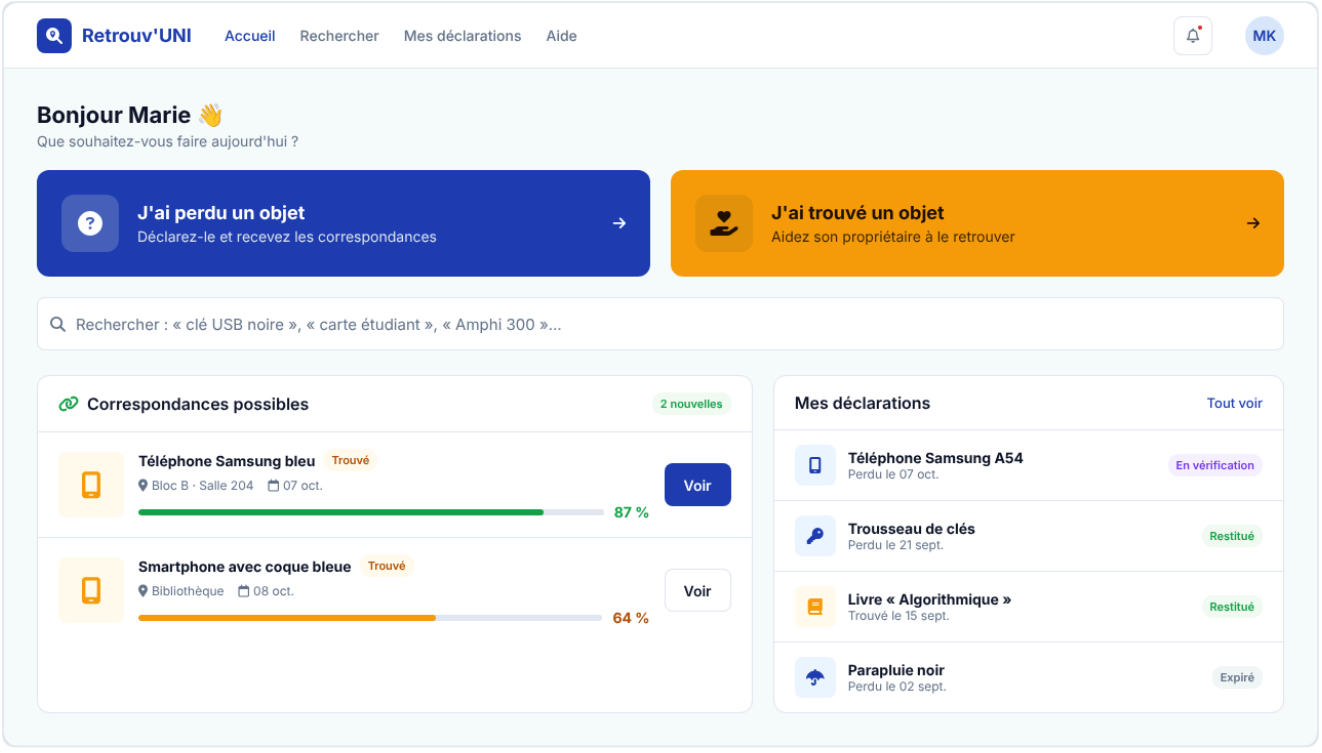
Importing the mockups into Figma

1. In Figma, install the **html.to.design** plugin (Plugins → search "html.to.design").
2. Import `maquettes/index.html` (hosted, e.g. via GitHub Pages, or pasted into the plugin's HTML import).
3. Each screen becomes a Figma frame with editable text, shape and auto-layout layers.
4. Create the colour and text styles from section 11, then convert buttons, inputs and badges into **components**.

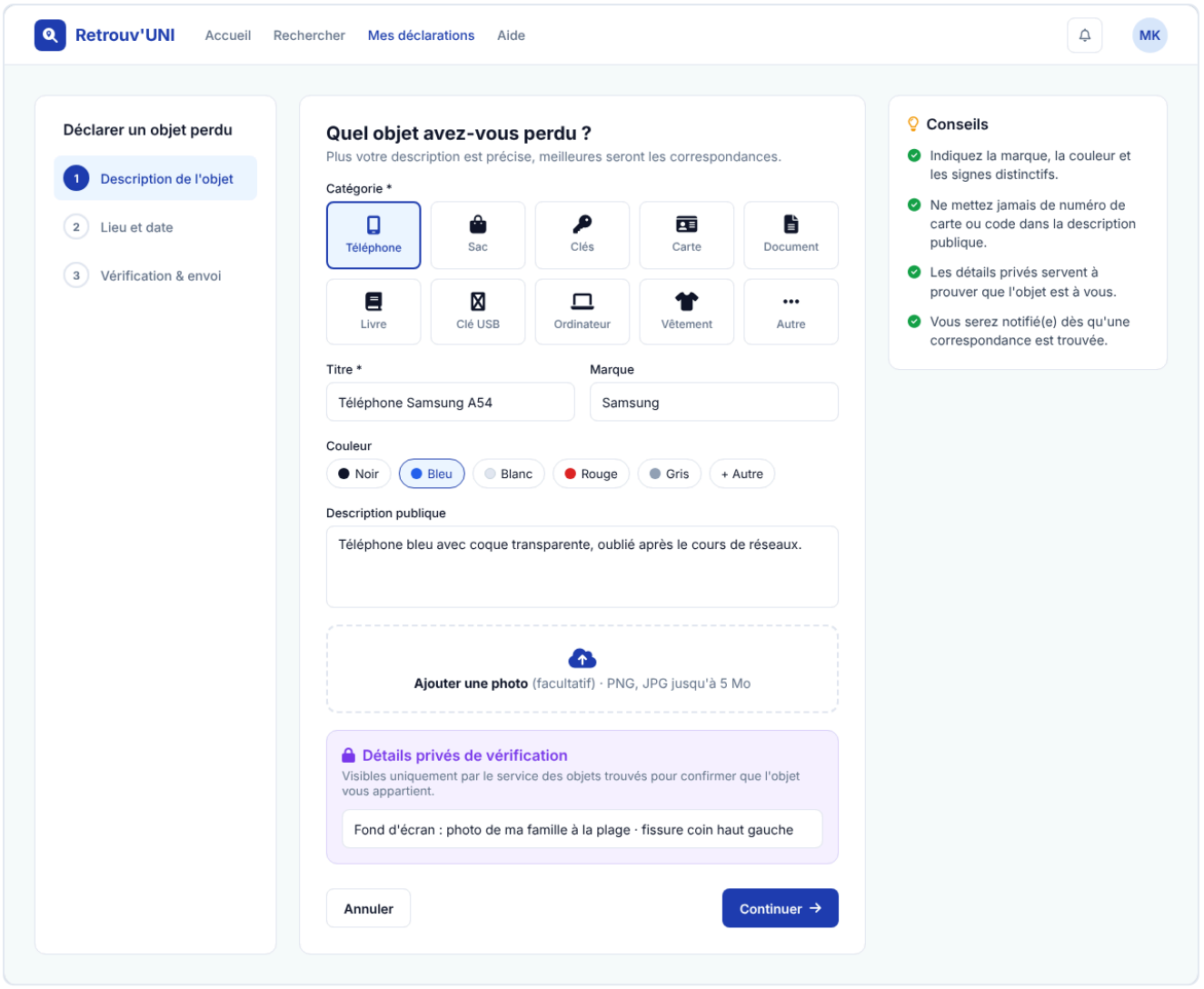
Mockup labels are currently in French; they will be localised to English (requirement NF-08).



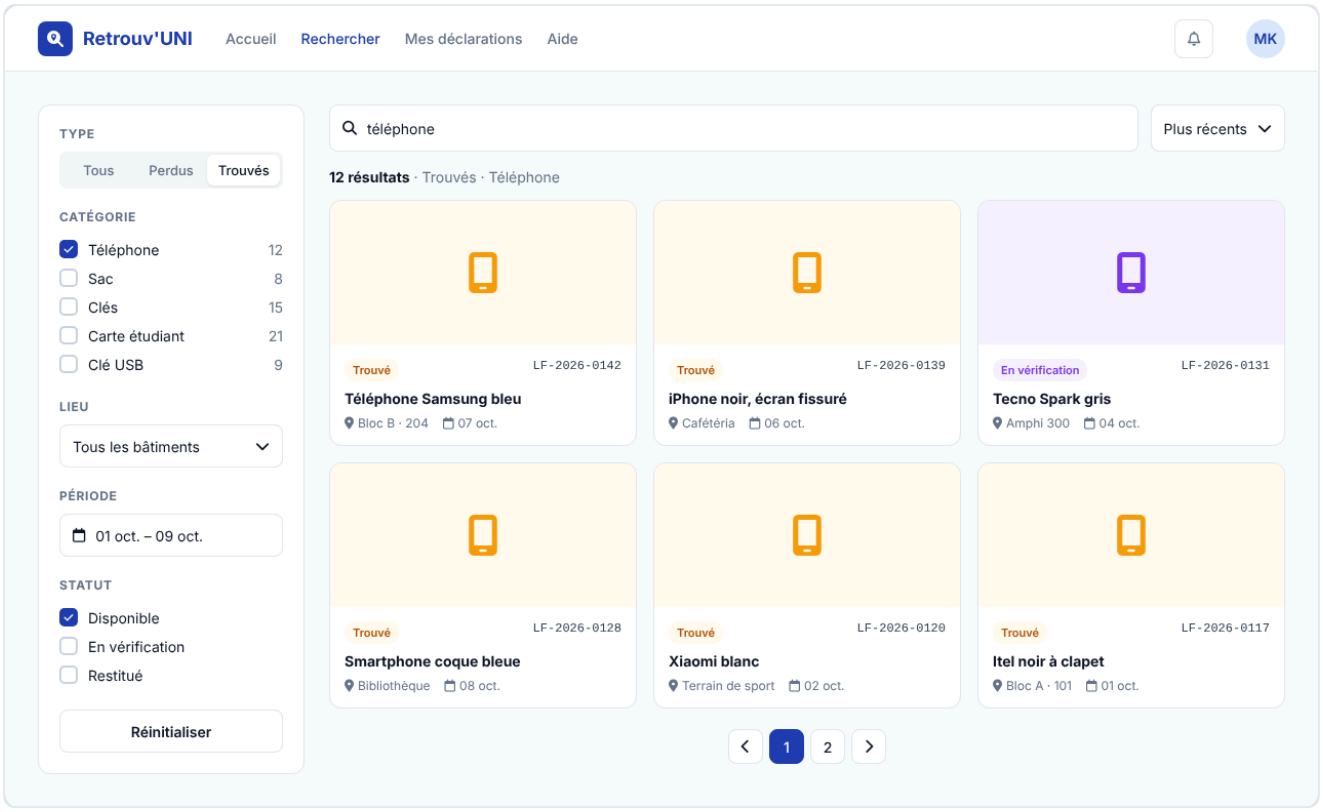
S1 — Login



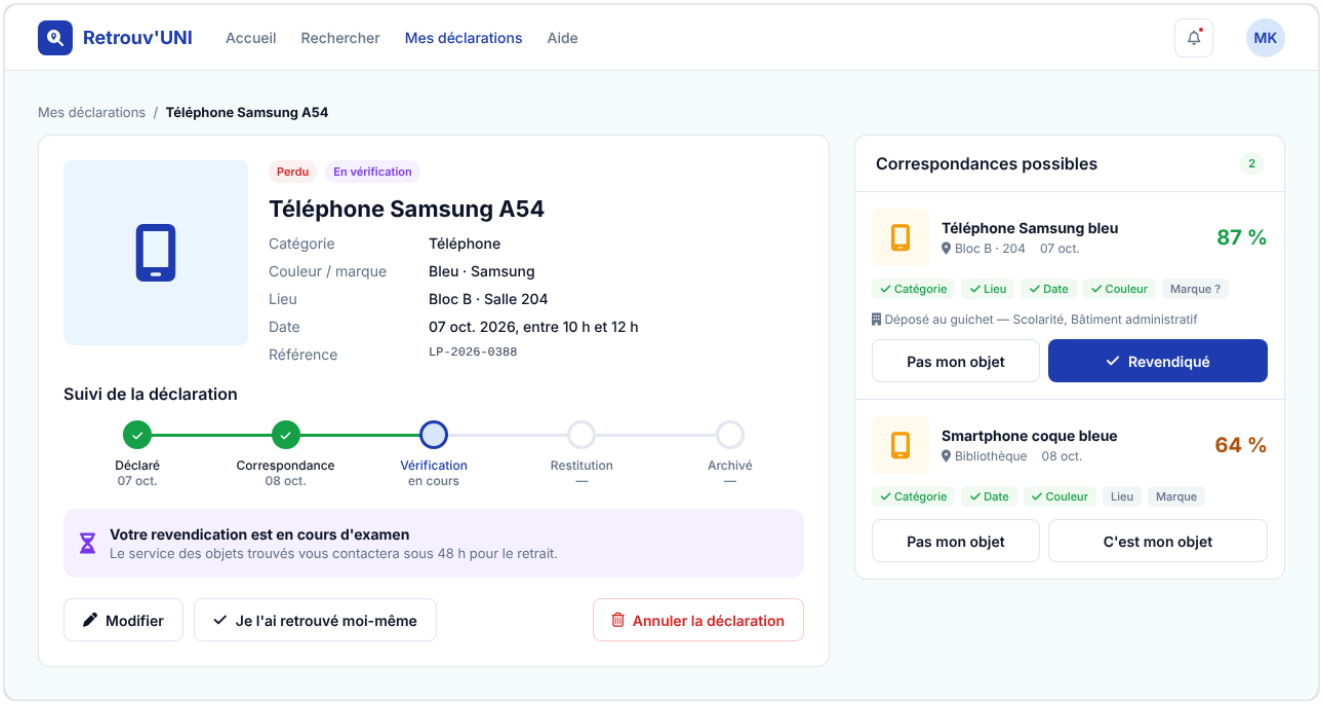
S2 — Student home / dashboard



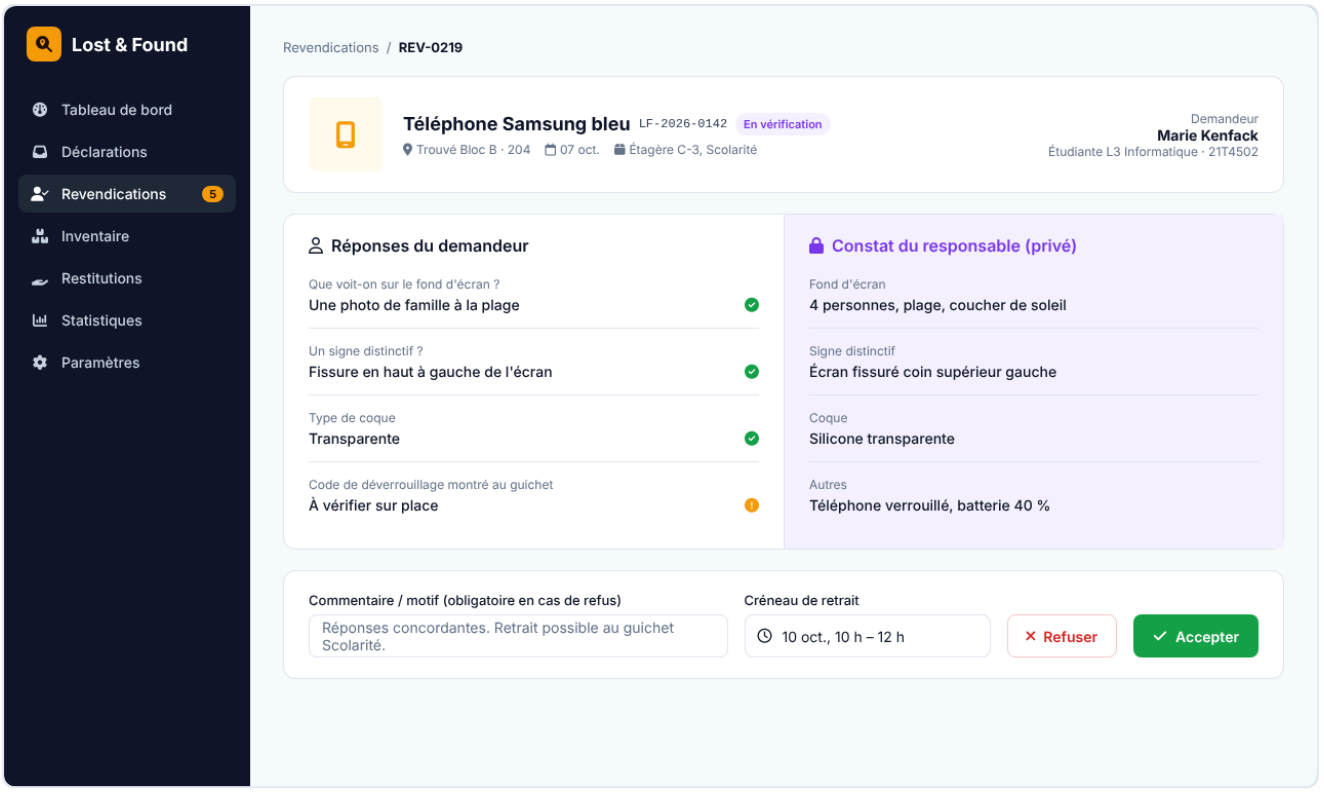
S3 — Report a lost item (step 1 of 3)



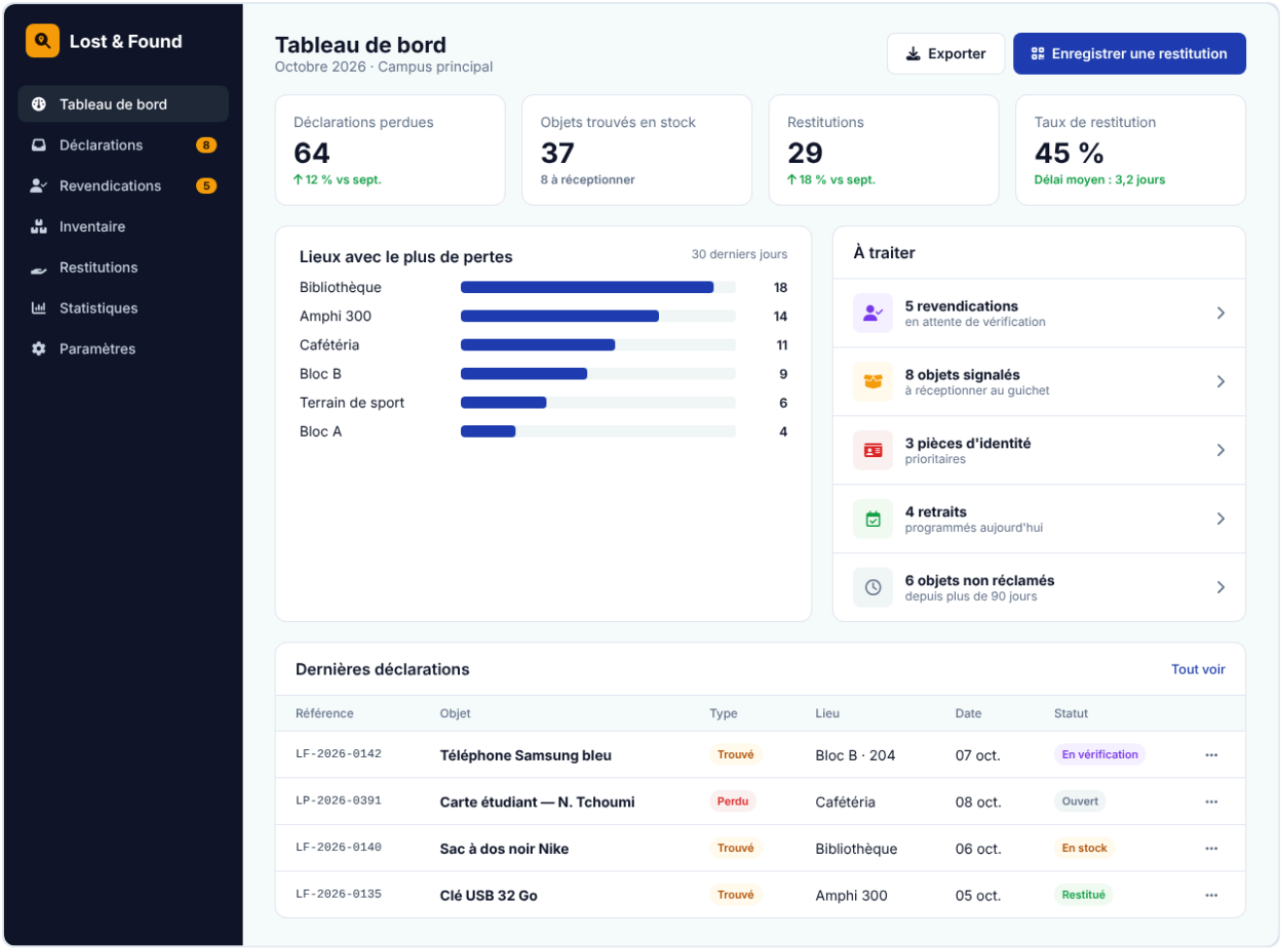
S4 — Search



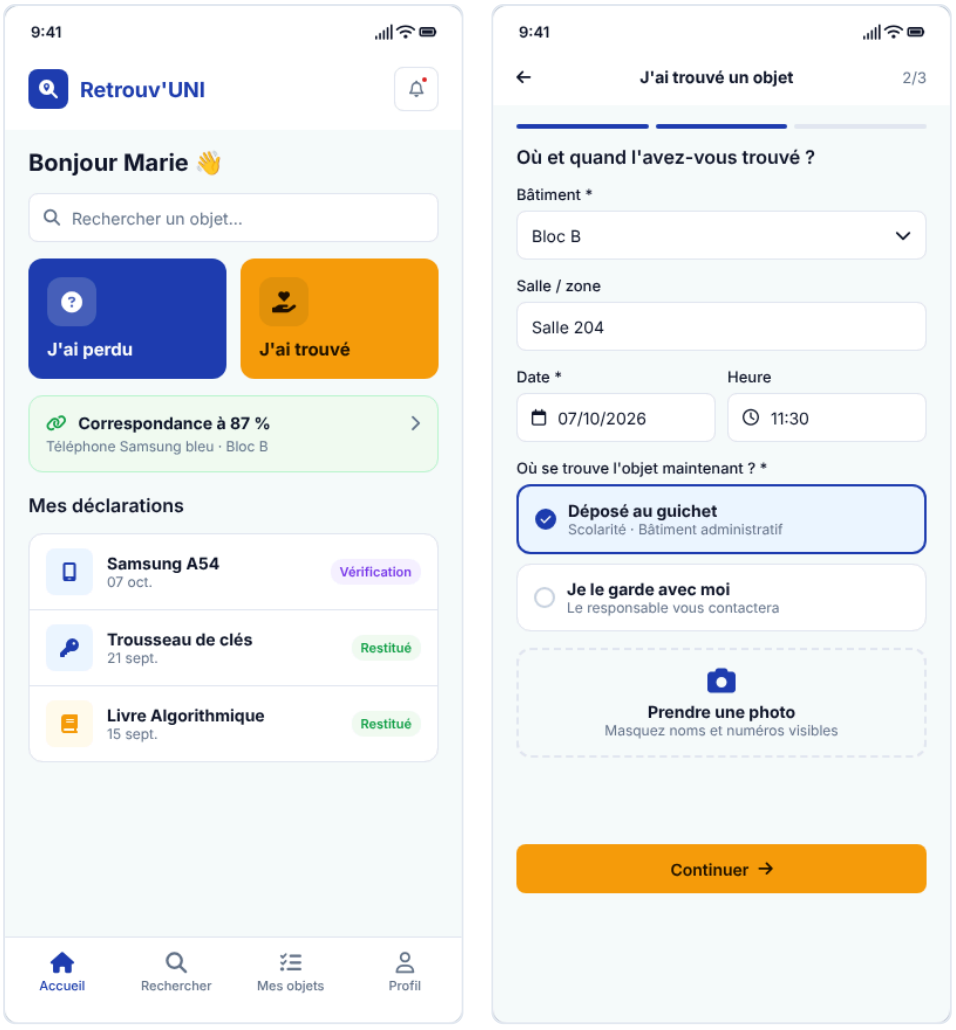
S5 — Report details and possible matches



S6 — Ownership verification (officer)



S7 — Officer dashboard



S8 — Mobile: home and found-item report

13. Provisional Schedule

Phase	Deliverables	Duration
1. Analysis	Requirements specification, feasibility study, business rules	2 weeks
2. Design	Use cases, conceptual/logical data model, architecture, Figma mockups	3 weeks
3. Development	Authentication, reports, search, matching, verification, return, administration	6 weeks
4. Testing	Unit, integration and user acceptance tests	2 weeks
5. Deployment	Go-live, user guide, final presentation	1 week

14. Acceptance Criteria

- A student can report a lost item in under 2 minutes from a phone.
- A matching found report triggers a match notification to the owner.
- No private field is visible to anyone other than the lost and found officer.
- A return is only possible after an accepted claim and the recording of an identity document.
- The officer can see the number of lost items per location and per month.

15. Expected Benefits

- Centralizes lost and found information.
- Makes it easier to search for missing belongings.
- Helps connect lost items with reported found items.
- Provides an organized ownership verification process.
- Keeps a history of reports and returned items.
- A structured process helps items reach their legitimate owners.
- The university can see which locations or periods have the most lost or found items.

16. Feasibility

The scope is defined around one workflow: **Report → Search/Match → Verify → Return → Record**. We do not promise to recover stolen items; we address the information-management problem of how a university can organize lost and found information and facilitate recovery.

This scope provides enough material for requirements analysis, a feasibility study, business rules, use cases, database design, system architecture, interface design, testing and deployment planning, using web technologies the team already masters.